Background: Asymptotic Freeness and Large Deviations

We start by recalling the basic asymptotic freeness result.

Proposition: Let $\left(A_1^N, \dots, A_r^N\right)$ be an independent r-tuple of GUE $N \times N$ matrices. Then, the family $A_1^N, \dots, A_r^N$ converge in distribution to r independent semicircular elements, $s_1, \dots, s_r \in B(\mathcal{F}(\mathbb{C}^r))$, in the sense that for all $m \geq 1$ and all $1 \leq i_1, \dots, i_m \leq r$, we have

$$\lim_{N\to\infty} E\left[\operatorname{tr}\left(A_{i_1}^N \cdots A_{i_m}^N\right)\right] = \varphi(s_{i_1} \cdots s_{i_m}),$$

where φ is the vacuum state, $\varphi(T) = \langle T\Omega, \Omega\rangle$.

In fact, this collection is *almost surely* asymptotically free, in the following sense. Suppose we have two random matrices A^N and B^N defined on probability spaces (X_N, μ_N). Define

$$\begin{aligned} X &\coloneqq \prod_{N\in\mathbb{N}} X_N \\ \mu &\coloneqq \prod_{N\in\mathbb{N}} \mu_N, \end{aligned}$$

where the latter is the product measure on X. The matrices A^N and B^N are said to be almost surely asymptotically free if there exists a noncommutative probability space (A, φ) and $a, b \in A$, and for almost all $x = (x_N)_N \in X$, we have $A^N(x_N), B^N(x_N) \in (\mathbb{M}_N, \operatorname{tr})$ converge in distribution to a, b.

Now, from here, we may ask a seemingly simple question: as N grows large, how likely are we to encounter other distributions? To make this sense more precise, we consider a random $N \times N$ self-adjoint matrix A, and let

$$\mu_A = \frac{1}{N}\sum_{i=1}^{N} \delta_{\lambda_1}$$

be its empirical spectral distribution. This is a random probability measure on $\mathbb{R}$, and as $N \to \infty$, the semicircle law gives that μ_A converges weakly to the semicircle distribution; this can be strengthened to almost sure convergence by an application of the argument for asymptotic freeness. The question then becomes, how quickly does the deviation between μ_A and any other probability distribution ν decrease as N increases (i.e., "how fast" does it converge)? This is where the theory of large deviations starts to take shape.

Much of this exposition related to the classical notions of entropy will be centered around results discussed in [MS17, Ch. 7].

Large Deviations for Random Variables

We start with one of the classical examples of convergence of random variables to introduce large deviations. Consider a sequence of independent and identically distributed real-valued random variables $(X_i)_i$ with common distribution μ. Set

$$\begin{aligned} S_n &= \frac{1}{n}\sum_{i=1}^{n} X_i. \\ m &= E[X_1] \\ v &= E\left[X_1^2\right] - m^2. \end{aligned}$$

Then, we have that if $E\left[X_1^2\right] < \infty$, the central limit theorem says that $S_n \approx m + \frac{\sigma}{\sqrt{n}} N(0,1)$.

If μ is the standard Gaussian distribution, then this gives that S_n is distributed as $N(0, 1/n)$; we then get that

$$P(S_n \in [x, x+dx]) \approx \sqrt{\frac{n}{2\pi}} e^{-nx^2} dx.$$

Asymptotically, this gives that the probability that S_n is near the value $x \in \mathbb{R}$ decays exponentially in n determined by a rate function $I(x) = x^2/2$.

We will now generalize this result. In particular, if we let μ be any distribution discussed above (rather than simply the normal distribution), then we will find a rate function $I(x)$ such that

$$e^{-nI(x)} \sim P(S_n > x)$$

whenever $x > m$, and whenever $x < m$

$$e^{-nI(x)} \sim P(S_n < x).$$

For a given distribution μ, we can compute the rate function by using a family of basic manipulations. If $x > m$, then for all $\lambda \geq 0$, we may use Markov's inequality to obtain

$$\begin{aligned}
P(S_n > x) &= P(nS_n > nx) \\
&= P\left(e^{\lambda(nS_n - nx)} \geq 1\right) \\
&\leq E\left[e^{\lambda(nS_n - nx)}\right] \\
&= e^{-\lambda nx} E\left[e^{\lambda(X_1 + \cdots + X_n)}\right] \\
&= \left(e^{-\lambda x} E\left[e^{\lambda X}\right]\right)^n,
\end{aligned}$$

where X is identically distributed to each of the X_i, and we use the fact that the X_i are independent. We may then define

$$\Lambda(\lambda) = \ln E\left[e^{\lambda X}\right] \tag{$*$}$$

to be an extended real-valued function, but we only consider μ for which $\Lambda(\lambda)$ is finite for all λ in an open neighborhood of 0. The equation $(*)$ is known as the cumulant generating function for μ.

This gives the inequality

$$P(S_n > x) \leq e^{-n(\lambda x - \Lambda(\lambda))}.$$

Since ln is a concave function, Jensen's inequality gives

$$\begin{aligned}
\Lambda(\lambda) &\geq E\left[\ln\left(e^{\lambda X}\right)\right] \\
&= E[\lambda X] \\
&= \lambda m.
\end{aligned}$$

In particular, for any $\lambda < 0$ and $x > m$, we have $-n(\lambda x - \Lambda(\lambda)) \geq 0$, meaning this equation is valid for all λ. In particular, we have

$$P(S_n > x) \leq \inf_{\lambda \in \mathbb{R}} e^{-n\lambda x - \Lambda(\lambda)}.$$

Now, we observe that Λ is convex. This follows from Hölder's inequality

$$E\left[e^{(1-t)\lambda_1 x + t\lambda_2 x}\right] \leq E\left[e^{\lambda_1 x}\right]^{1-t} E\left[e^{\lambda_2 x}\right]^t$$

so that

$$\Lambda((1-t)\lambda_1 + t\lambda_2) \leq (1-t)\Lambda(\lambda_1) + t\Lambda(\lambda_2).$$

Defining the *Legendre transform* of Λ by

$$\Lambda^*(x) = \sup_{\lambda \in \mathbb{R}} (\lambda x - \Lambda(\lambda)),$$

we find that this is a convex function of x, as it is a supremum of a family of convex functions of x.

Now, since $\Lambda(0) = 0$, it follows that $\Lambda^*(x) \geq 0$, and has $\Lambda^*(m) = 0$. In particular, this gives

$$P(S_n > x) \leq e^{-n\Lambda^*(x)}$$

whenever $x > m$.

It can also be shown that $e^{-n\Lambda^*(x)}$ is an asymptotic lower bound, in that

$$\liminf_{n\to\infty} \frac{1}{n} \ln P(x - \delta < S_n < x + \delta) \geq -\Lambda^*(x)$$

for all $x \in \mathbb{R}$ and all $\delta > 0$. The method for doing so is outlined in [MS17, Ch. 7, Section 2], and results in Cramér's theorem for real-valued random vectors.

Theorem (Cramér's Theorem)**:** Let $X_1, X_2, \dots$ be a sequence of independent and identically distributed random vectors in $\mathbb{R}^d$ with common distribution μ. Define

$$\begin{aligned} \Lambda(\lambda) &= \ln E\left[e^{\langle \lambda, X_i\rangle}\right] \\ \Lambda^*(x) &= \sup_{\Lambda\in\mathbb{R}^d} (\langle \lambda, x\rangle - \Lambda(\lambda)), \end{aligned}$$

and assume that $\Lambda(\lambda) < \infty$ for all λ. Set $S_n = \frac{1}{n}(X_1 + \cdots + X_n)$. Then, the distribution μ_{S_n} satisfies has

- $x \mapsto \Lambda^*(x)$ is convex;
- $\left\{x \in \mathbb{R}^d \;\middle|\; \Lambda^*(x) \leq \alpha\right\}$ is compact for all $\alpha \in \mathbb{R}$;
- for any closed $F \subseteq \mathbb{R}^d$,

$$\limsup_{n\to\infty} \frac{1}{n} \ln P(S_n \in F) \leq -\inf_{x\in F} \Lambda^*(x),$$

- and for any open $G \subseteq \mathbb{R}^d$,

$$\liminf_{n\to\infty} \frac{1}{n} \ln P(S_n \in G) \geq -\inf_{x\in G} \Lambda^*(x).$$

This theorem defines precisely the large deviation principle that the partial sums satisfy — namely, it is the Legendre transform of the cumulant-generating function.

Large Deviations for the Empirical Distribution

Now, our next step is to develop an analogous large deviation principle for the empirical distribution of the random variables $X_1, X_2, \dots$. This will give us the idea of classical entropy.

We start by considering the case of (independent and identically distributed) random variables $X_i \colon \Omega \to A$ taking values in a finite set $\{a_1, \dots, a_d\}$, and define $p_k := P(X_i = a_k)$. We expect that, as $n \to \infty$, the empirical distribution of the X_i should converge to the probability measure $(p_1, \dots, p_d)$ on A.

Toward this end, let $Y_i \colon \Omega \to \mathbb{R}^d$ be defined by

$$Y_i := \left(\chi_{\{a_1\}}(X_i), \dots, \chi_{\{a_d\}}(X_i)\right).$$

We observe that p_k is the probability that Y_i will have 1 in position k and 0 elsewhere, and that $\frac{1}{n}(Y_1 + \cdots + Y_n)$ gives the relative frequency of $a_1, \dots, a_d$ — i.e., this has the same information as the empirical distribution of $X_1, \dots, X_n$.

Any probability measure on A is a d-tuple of positive real numbers satisfying $q_1 + \cdots + q_d = 1$. By Cramér's theorem and our discussion above, we have

$$\begin{aligned} P\left(\frac{1}{n}(\delta_{X_1} + \cdots + \delta_{X_n}) \approx (q_1, \dots, q_d)\right) &= P\left(\frac{1}{n}(Y_1 + \cdots + Y_n) \approx (q_1, \dots, q_d)\right) \\ &\sim e^{-n\Lambda^*(q_1,\dots,q_d)}. \end{aligned}$$

Applying our definitions for Λ and Λ^*, we have

$$\begin{aligned} \Lambda(\lambda_1, \dots, \lambda_d) &= \ln\left(p_1 e^{\lambda_1} + \cdots + p_d e^{\lambda_d}\right) \\ \Lambda^*(q_1, \dots, q_d) &= \sup_{(\lambda_1,\dots,\lambda_d)} \left(\lambda_1 q_1 + \cdots + \lambda_d q_d - \ln\left(p_1 e^{\lambda_1} + \cdots + p_d e^{\lambda_d}\right)\right). \end{aligned}$$

To compute the supremum over all tuples, we find that the partial derivatives with respect to each λ_i are given by

$$q_i - \frac{1}{p_1 e^{\lambda_1} + \cdots + p_d e^{\lambda_d}} p_i e^{\lambda_i},$$

so by concavity, we get that the maximum value occurs when

$$\lambda_i = \ln\left(\frac{q_i}{p_i}\right) + \Lambda(\lambda_1, \dots, \lambda_d).$$

We thus get

$$\Lambda^*(q_1, \dots, q_d) = \sum_{i=1}^{d} q_i \ln\left(\frac{q_i}{p_i}\right).$$

The quantity on the right is the relative Shannon entropy $H((q_1, \dots, q_d)|(p_1, \dots, p_d))$, which is strictly positive except when $q_1 = p_1, \dots, q_d = p_d$.

In particular, we have that $(p_1, \dots, p_d)$ admits a large deviation principle with rate function given by the relative Shannon entropy. That this holds for any distribution is known as Sanov's theorem.

Theorem (Sanov's Theorem)**:** Let $X_1, X_2, \dots$ be a sequence of independent and identically distributed real random variables with common distribution μ, and let

$$\nu_n := \frac{1}{n} \sum_{i=1}^{n} \delta_{X_i}$$

be the empirical distribution. Then, the family $\{\nu_n\}_{n \geq 1}$ satisfies a large deviation principle given by the rate function

$$I(\nu) = \begin{cases} \int p \ln(p)\, d\mu & d\nu = p\, d\mu \\ +\infty & \text{else.} \end{cases}$$

One-Dimensional Free Entropy: A Heuristic Approach

The next logical step after defining a large deviation principle for a sequence of random variables is to define such a quantity for single free random variables.

In [Voi93], Voiculescu used a heuristic method based on random matrix theory to establish the value $\Sigma(x)$ for some random variable x in a C^*-probability space $(\mathcal{A}, \varphi)$ distributed according to a compactly supported measure ν on $\mathbb{R}$ (i.e., its spectral measure with respect to the state φ).

It can be established (as in [MS17, Exercise 1.8]) that if X is a $N \times N$ GUE matrix, then the density inside the real vector space $\mathbb{M}_N(\mathbb{C})_{\text{s.a.}}$ is given by

$$dP_N(X) = K e^{-N^2/2 \operatorname{tr}(X^2)}\, dm,$$

where m is the Lebesgue measure on $\mathbb{R}^{N^2}$ and tr denotes the normalized trace. It can be shown, as in [AGZ10, Theorem 2.5.2], that the joint distribution of the eigenvalues $\lambda_1(X) \leq \cdots \leq \lambda_N(X)$ is absolutely continuous with respect to the Lebesgue measure and has density given by

$$dQ_N(\lambda_1, \dots, \lambda_N) = \frac{N^{N^2/2}}{(2\pi)^{N/2} \prod_{j=1}^N j!} e^{-\left(N \sum_{i=1}^N \lambda_i^2\right)/2} \prod_{i<j} (\lambda_i - \lambda_j)^2 \prod_{i=1}^N d\lambda_i$$

Heuristically, letting μ_A denote the empirical spectral distribution on $\mathbb{M}_N(\mathbb{C})_{\text{s.a.}}$ given by

$$\mu_A = \frac{1}{N}\left(\delta_{\lambda_1(A)} + \cdots + \delta_{\lambda_N(A)}\right),$$

we may consider a large deviation principle with $P_N(\mu_A \approx \nu) \sim e^{-N^2 I(\nu)}$, and write it out as

$$\begin{aligned} P_N(\mu_A \approx \nu) &= Q_N\left(\frac{1}{N}(\delta_{\lambda_1} + \cdots + \delta_{\lambda_N})\right) \\ &= \frac{N^{N^2/2}}{(2\pi)^{N/2} \prod_{j=1}^N j!} \int_E e^{-\left(N \sum_{i=1}^N \lambda_i^2\right)/2} \prod_{i<j} (\lambda_i - \lambda_j)^2 \prod_{i=1}^N d\lambda_i, \end{aligned}$$

where E is the given set in the Q_N. Whenever $\frac{1}{N}\left(\delta_{\lambda_1(A)} + \cdots + \delta_{\lambda_N(A)}\right) \approx \nu$, we have that

$$-\frac{N}{2} \sum_{i=1^N} \lambda_i^2 = -\frac{N^2}{2}\left(\frac{1}{N}\sum_{i=1}^N \lambda_i^2\right)$$

is a Riemann sum for the integral of t^2 with respect to ν. Furthermore, we have

$$\prod_{i<j} (\lambda_i - \lambda_j)^2 = e^{\sum_{i \neq j} \ln|\lambda_i - \lambda_j|}$$

The sum inside the argument of the exponential is a Riemann sum for the quantity $N^2 \iint \ln|s-t| d\nu(s) d\nu(t)$. Combining all these findings together, we get the following large deviation principle, which was proven rigorously in [BAG97] for the general case of Gaussian random matrices with Dyson index β.

Theorem: Let

$$I(\nu) = -\iint \ln|s-t| \, d\nu(s)\nu(t) + \frac{1}{2}\int t^2 \, d\nu(t) - \frac{3}{4}.$$

Then, the following hold.

(i) The function $I \colon \mathcal{M} \to [0, \infty]$ is a well-defined convex function with compact level sets on the space of probability measures on $\mathbb{R}$ that has minimum value 0 at Wigner's semicircle law.

(ii) The empirical spectral distribution satisfies a large deviation principle with respect to Q_N with rate function I. That is,

$$\liminf_{N\to\infty} \frac{1}{N^2} \ln Q_N\left(\frac{1}{N}(\delta_{\lambda_1} + \cdots + \delta_{\lambda_N}) \in G\right) \geq -\inf_{\nu \in G} I(\nu)$$

for any open $G \subseteq \mathcal{M}$, and

$$\limsup_{N\to\infty} \frac{1}{N^2} \ln Q_N\left(\frac{1}{N}(\delta_{\lambda_1} + \cdots + \delta_{\lambda_N}) \in F\right) \leq -\inf_{\nu \in F} I(\nu)$$

for any closed $F \subseteq \mathcal{M}$.

Microstates Free Entropy

Voiculescu's original definition of free entropy in the case with more than one free random variable can be viewed as a generalization of the microstates approach to entropy for a classical discrete random variable, discussed in the survey [Voi01].

Consider a discrete random variable with output values in $\{1, \dots, n\}$ assigned with probabilities $p_1, \dots, p_n$. The microstates of this discrete random variable for a fixed N are then the set

$$\{f \mid f\colon \{1, \dots, N\} \to \{1, \dots, n\}\},$$

and for a fixed ε, the microstates that approximate this distribution are those f with

$$\left| \frac{\left|f^{-1}(\{j\})\right|}{N} - p_j \right| < \varepsilon,$$

where $|\cdot|$ denotes the (necessarily finite) cardinality. We denote the collection of such f by $\Gamma(p_1, \dots, p_n; \varepsilon, N)$. One may find the Shannon entropy by evaluating the limit

$$\lim_{\varepsilon \to 0} \lim_{N \to \infty} \frac{1}{N} \ln|\Gamma(p_1, \dots, p_n; \varepsilon, N)|$$

Understanding the Microstates Free Entropy

Instead of a classical random variable, we will let (M, τ) be a tracial von Neumann algebra,[I] and let $(x_1, \dots, x_n)$ be a tuple of self-adjoint free random variables in M.

For a fixed R, we let the set of microstates $\Gamma_R(x_1, \dots, x_n; m, k, \varepsilon)$ admit three degrees of approximation: m denotes the level of approximation of mixed moments, k denotes the size of the (self-adjoint) approximation matrices that have operator norm at most R, and ε denotes the closeness of the approximation. Put into symbols, we select all n-tuples $(A_1, \dots, A_n) \in (\mathbb{M}_k(\mathbb{C})_{\text{s.a.}})^n$ with each $\|A_j\| < R$ satisfying

$$\left|\tau\left(x_{i_1} \cdots x_{i_p}\right) - \operatorname{tr}_k\left(A_{i_1} \cdots A_{i_p}\right)\right| < \varepsilon$$

for all $1 \leq p \leq m$ and all multi-indices $\mathbf{i}\colon \{1, \dots, p\} \to \{1, \dots, n\}$. Here, tr_k is the normalized trace.

As in the case of entropy for classical random variables, free entropy emerges from a certain large deviation principle. Specifically, recall that the distribution of a family of noncommutative random variables is given by the collection of mixed moments with respect to the trace,

$$\Delta(x_1, \dots, x_n) = \left\{\tau\left(x_{i_1} \cdots x_{i_p}\right) \;\middle|\; p \in \mathbb{N}, \mathbf{i}\colon \{1, \dots, p\} \to \{1, \dots, n\}\right\}.$$

If we have an n-tuple of independent GUE matrices $(A_1, \dots, A_n) \in (\mathbb{M}_k(\mathbb{C})_{\text{s.a.}})^n$, we know that as $k \to \infty$, that there is almost sure convergence in distribution to a family $(s_1, \dots, s_n)$ of free semicirculars in M. The large deviations will then be given by

$$P_N(\Delta(A_1, \dots, A_n) \approx \Delta(x_1, \dots, x_n)) \sim e^{-k^2 I(x_1, \dots, x_n)},$$

where I is the free entropy.

To compute this value, we let λ denote the Lebesgue measure on $(\mathbb{M}_k(\mathbb{C})_{\text{s.a.}})^n \cong \mathbb{R}^{nk^2}$, and define

$$\chi(x_1, \dots, x_n) = \sup_{R>0} \inf_{m \in \mathbb{N}} \inf_{\varepsilon > 0} \limsup_{k \to \infty} \left(\frac{1}{k^2} \ln \lambda(\Gamma_R(x_1, \dots, x_n; m, k, \varepsilon)) + \frac{n}{2} \ln k \right).$$

It turns out that the value of R has minor influence, and we only need choose a fixed R greater than the norm of each x_i.

[I] A von Neumann algebra equipped with a faithful (injective on positive elements), normal (w^*-continuous), tracial ($\tau(xy) = \tau(yx)$) state ($\tau(1) = 1$) $\tau\colon M \to \mathbb{C}$.

Proposition ([Voi94, Proposition 2.2])**:** Let $C^2 = \tau\left(x_1^2 + \cdots + x_n^2\right)$. Then,

$$\chi(x_1, \dots, x_n) \leq 2^{-1} n \ln\left(2\pi e n^{-1} C^2\right).$$

Proof. We will instead prove a slightly different inequality. Define

$$\chi_R(x_1, \dots, x_n; m, k, \varepsilon) = \ln(\lambda(\Gamma_R(x_1, \dots, x_n; m, k, \varepsilon))),$$

and we will show that

$$\chi_R(x_1, \dots, x_n; m, k, \varepsilon) \leq 2^{-1} n k^2 \left(\ln\left(2\pi e n^{-1}\left(C^2 + n\varepsilon\right)\right) - \ln k\right),$$

assuming that $m \geq 2$. Applying the operations $\limsup_{k\to\infty}$, $\inf_{\varepsilon>0}$, $\inf_{m\in\mathbb{N}}$, and $\sup_{R>0}$ in succession will give us the free entropy.

For this, we use the p-dimensional Shannon entropy inequality

$$-\int f \ln(f)\, d\lambda_p \leq 2^{-1} p \ln\left(2\pi e p^{-1} a^2\right),$$

where f is some probability density function on $\mathbb{R}^p$ and

$$a^2 = \int \left(x_1^2 + \cdots + x_p^2\right) f\, d\lambda_p,$$

where λ_p denotes the Lebesgue measure on $\mathbb{R}^p$.

We apply this to the special case of the Lebesgue measure on $\left(\mathbb{M}_k(\mathbb{C})_{\text{s.a.}}\right)^n$. For this, we use the Hilbert–Schmidt metric

$$\langle (A_1, \dots, A_n), (B_1, \dots, B_n) \rangle = \sum_{i=1}^{n} \operatorname{tr}(A_i B_i),$$

and define the measure λ on $\mathbb{R}^{k^2}$ by pulling back the traditional Lebesgue measure on $\mathbb{R}^{k^2}$ under the map taking $\left\{ A \in \mathbb{M}_k(\mathbb{C})_{\text{s.a.}} \;\middle|\; \operatorname{Tr}\left(A^2\right) \leq R^2 \right\}$ to the closed ball of radius R^2 in $\mathbb{R}^{k^2}$.

Taking the indicator function in the k^2-dimensional Shannon entropy

$$f(A_1, \dots, A_n) = \begin{cases} 0 & (A_1, \dots, A_n) \notin \Gamma_R(x_1, \dots, x_n; m, k, \varepsilon) \\ (\lambda(\Gamma_R(x_1, \dots, x_n; m, k, \varepsilon)))^{-1} & (A_1, \dots, A_n) \in \Gamma_R(x_1, \dots, x_n; m, k, \varepsilon) \end{cases}$$

gives

$$\chi_R(x_1, \dots, x_n; m, k, \varepsilon) \leq 2^{-1} n k^2 \left(\ln\left(2\pi e n^{-1} k^{-2} a^2\right) - \ln k\right).$$

Finally, by the definition of the microstate space, we have

$$\left| \frac{1}{k} \int \operatorname{Tr}\left(\sum_{j=1}^{n} A_j^2\right) f\, d\lambda - \tau\left(\sum_{j=1}^{n} x_j^2\right) \right| < n\varepsilon,$$

meaning that $a^2 \leq C^2 + n\varepsilon$. □

In particular, this means that $\chi(x_1, \dots, x_n)$ is either finite or equal to $-\infty$. We will have more to discuss about the structural properties of M in these cases. There are more properties of the free entropy that have shown themselves to be useful. For instance, it can be shown that the free entropy is subadditive, in that

$$\chi(x_1, \dots, x_m, x_{m+1}, \dots, x_{m+n}) \leq \chi(x_1, \dots, x_m) + \chi(x_{m+1}, \dots, x_{m+n}).$$

In the one-variable case, the definition of the free entropy reduces to the definition in the definition from the heuristic approach, as it can be shown that the limsup is in fact a limit; that is, we have

$$\chi(x) = \int \ln|s-t|\, d\mu(s)d\mu(t) + \frac{3}{4} + \frac{1}{2}\ln(2\pi),$$

where μ denotes the distribution of x (i.e., the spectral measure of the image of x under the GNS representation of M with respect to the state τ).

The Microstates Free Entropy Dimension

The free entropy is a normalized limits of the logarithm of the volume of some collection of microstates; similarly, we can consider a normalized dimension, known as the *free entropy dimension* of a family $x_1, \dots, x_n \in M$, given by

$$\delta(x_1, \dots, x_n) = n + \limsup_{\varepsilon \to 0} \frac{\chi(x_1 + \varepsilon s_1, \dots, x_n + \varepsilon s_n)}{|\ln(\varepsilon)|},$$

where $s_1, \dots, s_n$ are a family of freely independent semicirculars that is also freely independent from $\{x_1, \dots, x_n\}$.

In most cases, it is useful to consider a modification to the free entropy dimension that involves the concept of entropy with respect to some family. If $x_1, \dots, x_n, y_1, \dots, y_r \in M$, $m, k \in \mathbb{N}$, and $\varepsilon > 0$, then we define the quantity

$$\Gamma_R(x_1, \dots, x_n \colon y_1, \dots, y_r; m, k, \varepsilon)$$

to be the collection of all matrices $(A_1, \dots, A_n) \in (\mathbb{M}_k(\mathbb{C})_{\text{s.a.}})^n$ such that there exists $(B_1, \dots, B_r) \in (\mathbb{M}_k(\mathbb{C})_{\text{s.a.}})^r$ satisfying

$$(A_1, \dots, A_n, B_1, \dots, B_r) \in \Gamma_R(x_1, \dots, x_n, y_1, \dots, y_r; m, k, \varepsilon).$$

Taking the requisite limsup with respect to k, infimum over all $\varepsilon > 0$, infimum over all $m \in \mathbb{N}$, and supremum over all $R > 0$ gives us the quantity $\chi(x_1, \dots, x_n \colon y_1, \dots, y_r)$. This allows us to define the modified free entropy dimension by

$$\delta_0(x_1, \dots, x_n) = n + \limsup_{\varepsilon \to 0} \frac{\chi(x_1 + \varepsilon s_1, \dots, x_n + \varepsilon s_n \colon s_1, \dots, s_n)}{|\ln(\varepsilon)|}$$

In [Voi96], some of the following properties of the modified free entropy dimension are discussed below:

- $\delta_0(a_1, \dots, a_n) \leq \delta_0(a_1, \dots, a_j) + \delta_0(a_j, \dots, a_n)$;
- if the family $a_1, \dots, a_n$ are freely independent, then $\delta_0(a_1, \dots, a_n) = n$;
- for any $a \in M_{\text{s.a.}}$, we have

  $$\delta_0(a) = 1 - \sum_{t \in \sigma(a)} (\mu(\{t\}))^2,$$

  where μ is the distribution of τ;
- if $\chi(a_1, \dots, a_n) > -\infty$, then $\delta_0(a_1, \dots, a_n) = n$.

It can be observe that δ_0 is a noncommutative generalization of Minkowski dimension in a certain sense, as is discussed in [Jun02]. If $A \subseteq \mathbb{R}^d$ is bounded, then we may define $N_\varepsilon(A)$ to be the set

$$N_\varepsilon(A) = \left\{ x \in \mathbb{R}^d \;\middle|\; \operatorname{dist}_A(x) < \varepsilon \right\}.$$

The Minkowski dimension of A is then given by the quantity $d + \limsup_{\varepsilon \to 0} \frac{\ln \lambda(N_\varepsilon(A))}{|\ln(\varepsilon)|}$, where λ denotes the Lebesgue measure on $\mathbb{R}^d$.

Jung later proved that δ_0 can also be considered in the context of a type of packing dimension. In $\mathbb{R}^d$, the packing dimension is defined as follows. If A is as above, then we can define $P_\varepsilon(A)$ to be the maximum number of elements in a collection of mutually disjoint open balls of radius ε in A, while $K_\varepsilon(A)$ denotes the minimum number of open balls of radius ε necessary to cover A. It can be shown then that

$$d + \limsup_{\varepsilon\to 0} \frac{\ln \lambda(N_\varepsilon(A))}{|\ln(\varepsilon)|} = \limsup_{\varepsilon\to 0} \frac{P_\varepsilon(A)}{|\ln(\varepsilon)|} = \limsup_{\varepsilon\to 0} \frac{K_\varepsilon(A)}{|\ln(\varepsilon)|}.$$

One benefit of the approach regarding packing numbers is that it does not invoke any kind of measure-theoretic structure on $\mathbb{R}^d$, purely using the Euclidean metric structure. While Voiculescu's microstate structure used the Hilbert–Schmidt inner product and the isomorphism between $\mathbb{R}^{k^2} \cong \mathbb{M}_k(\mathbb{C})_{\text{s.a.}}$ to construct the notion of the Lebesgue measure on space of self-adjoint matrices, this method should allow us to directly use the Hilbert–Schmidt inner product to construct the free entropy dimension without invoking the Lebesgue measure.

For this, given a (compact) metric space (X, d), define $P_\varepsilon(X, d)$ to be the maximum number of disjoint open balls contained in X, and let $K_\varepsilon(X, d)$ be the minimum number of open balls necessary to cover X. We have the inequality $P_\varepsilon(X, d) \geq K_{2\varepsilon}(X, d) \geq P_{4\varepsilon}(X, d)$. It is then possible to define

$$\begin{aligned}
\mathbb{P}_\varepsilon(a_1, \dots, a_n) &= \sup_{R>0} \inf_{m\in\mathbb{N}} \inf_{\gamma>0} \limsup_{k\to\infty} \left(\frac{1}{k^2} P_\varepsilon(\Gamma_R(a_1, \dots, a_n; m, k, \gamma), \rho_k) \right) \\
\mathbb{K}_\varepsilon(a_1, \dots, a_n) &= \sup_{R>0} \inf_{m\in\mathbb{N}} \inf_{\gamma>0} \limsup_{k\to\infty} \left(\frac{1}{k^2} K_\varepsilon(\Gamma_R(a_1, \dots, a_n; m, k, \gamma), \rho_k) \right),
\end{aligned}$$

where ρ_k denotes the Hilbert–Schmidt metric on $(\mathbb{M}_k(\mathbb{C})_{\text{s.a.}})^n$ induced by the inner product

$$\langle (A_1, \dots, A_n), (B_1, \dots, B_n) \rangle = \sum_{i=1}^{n} \operatorname{Tr}(A_i B_i)$$

As with the case of traditional Minkowski dimension, it is possible to show that the modified free entropy dimension can be considered with regard to packing numbers. The following is a special case of [Jun02, Lemma 2.2]

Corollary: Suppose $\{a_1, \dots, a_n\}$ is a set of elements of (M, τ), and $\{s_1, \dots, s_n\}$ is a set of freely independent semicirculars in M that is also freely independent from $\{a_1, \dots, a_n\}$. Then,

$$\begin{aligned}
\delta_0(a_1, \dots, a_n) &= n + \limsup_{\varepsilon\to 0} \frac{\chi(a_1 + \varepsilon s_1, \dots, a_n + \varepsilon s_n \colon s_1, \dots, s_n)}{|\ln(\varepsilon)|} \\
&= \limsup_{\varepsilon\to 0} \frac{\mathbb{K}_\varepsilon(a_1, \dots, a_n)}{|\ln(\varepsilon)|} \\
&= \limsup_{\varepsilon\to 0} \frac{\mathbb{P}_\varepsilon(a_1, \dots, a_n)}{|\ln(\varepsilon)|}.
\end{aligned}$$

Applications: Structural Properties of Free Group Factors

The primary hope with the value χ is that it will provide for some kind of invariant of a von Neumann algebra, and in particular, structural properties of the free group factors $L(\mathbb{F}_r)$.

Recall that if Γ is any countable discrete group, the group von Neumann algebra of Γ, denoted $L(\Gamma)$, is defined via the left regular representation of Γ as follows. If $\ell_2(\Gamma)$ denotes the square-summable sequences indexed by Γ, then for any $s \in \Gamma$, the unitary map $\lambda_s \in B(\ell_2(\Gamma))$ is defined by $\lambda_s f(t) = f\left(s^{-1}t\right)$. The von Neumann subalgebra of $B(\ell_2(\Gamma))$ denoted by these unitaries is the group von Neumann algebra.

The most important fact about the group von Neumann algebra is that if Γ has infinite conjugacy classes for any non-identity element $g \in \Gamma$, then $L(\Gamma)$ is a II_1 factor; that is, a von Neumann algebra with trivial center, no nonzero abelian projections, and is equipped with a trace given by

$$\tau(\lambda_s) = \langle \lambda_s \delta_e, \delta_e \rangle,$$

where δ_e is the element of the orthonormal basis of $\ell_2(\Gamma)$ corresponding to the identity.

We call the von Neumann algebras $L(\mathbb{F}_r)$ generated by the free groups *free group factors*. The free group factor isomorphism problem asks if, for any $r \neq s$, we have $L(\mathbb{F}_r) \ncong L(\mathbb{F}_s)$. In [PV80], it was shown that the reduced group C^*-algebras[II] $C^*_\lambda(\mathbb{F}_r)$ and $C^*_\lambda(\mathbb{F}_s)$ are not isomorphic if $r \neq s$, via a calculation of certain K-theoretic invariant.

Voiculescu developed the concepts of free entropy and free entropy dimension in order to hopefully solve the free group factor isomorphism problem. Yet, in the process, it was shown that there were many useful properties that could be shown via the use of free entropy.

Definition: If $N \leq M$ is an inclusion of von Neumann algebras, then the normalizer of N is the von Neumann subalgebra of M generated by the unitaries $u \in M$ with $uNu^* = N$.

We say N is maximal abelian if it is abelian and not contained in any abelian subalgebra. A maximal abelian subalgebra $N \leq M$ is called a *Cartan subalgebra* if its normalizer generates M.

In [Voi96], it was shown that if there is a generating tuple $(x_1, \dots, x_n)$ for a von Neumann algebra M such that $\chi(x_1, \dots, x_n) > -\infty$, then M does not have a Cartan subalgebra. This resolved a long-standing conjecture that every II_1 factor contains a Cartan subalgebra in the negative.

We start by showing that a free group factor $L(\mathbb{F}_r)$ is generated by a family $\{s_1, \dots, s_r\}$ of r freely independent semicirculars.

Definition: If (M, φ) is a W^*-probability space with φ a faithful normal state, then an element $u \in M$ is called a *Haar unitary* if

$$\varphi(u^n) = 0$$

for all $n \neq 0$.

Theorem ([HP00, Proposition 7.1.1])**:** Let (M, φ) be a W^*-probability space with faithful normal state φ. If $\{s_1, \dots, s_r\}$ is a family of freely independent semicirculars in M, then the von Neumann algebra $\{s_1, \dots, s_r\}''$ is isomorphic to $L(\mathbb{F}_r)$.

Proof. We observe that if $a \in M_{\text{s.a.}}$ has a $(0,1)$ semicircular distribution, then the function $f\colon [-2,2] \to [0,1]$ given by $df = \frac{1}{2\pi}\sqrt{4-t^2}\,dt$ allows us to define a Haar unitary $u = e^{2\pi i f(a)}$, so we will show an equivalent condition. Namely, if $\{u_1, \dots, u_r\}$ is a family of r freely independent Haar unitaries, then $\{u_1, \dots, u_r\}'' \cong L(\mathbb{F}_r)$. □

References

[PV80] M. Pimsner and D. Voiculescu. "Exact sequences for K-groups and Ext-groups of certain cross-product C^*-algebras". In: *J. Operator Theory* 4.1 (1980), pp. 93–118. ISSN: 0379-4024.

[Voi93] Dan Voiculescu. "The analogues of entropy and of Fisher's information measure in free probability theory. I". In: *Comm. Math. Phys.* 155.1 (1993), pp. 71–92. ISSN: 0010-3616,1432-0916. URL: http://projecteuclid.org/euclid.cmp/1104253200.

[Voi94] Dan Voiculescu. "The analogues of entropy and of Fisher's information measure in free probability theory. II". In: *Invent. Math.* 118.3 (1994), pp. 411–440. ISSN: 0020-9910,1432-1297. DOI: 10.1007/BF01231539. URL: https://doi.org/10.1007/BF01231539.

[Voi96] D. Voiculescu. "The Analogues of Entropy and of Fisher's Information Measure in Free Probability Theory III: The Absence of Cartan Subalgebras". In: *Geometric and functional analysis* 6.1 (1996), pp. 172–200. URL: http://eudml.org/doc/58223.

[BAG97] G. Ben Arous and A. Guionnet. "Large deviations for Wigner's law and Voiculescu's non-commutative entropy". In: *Probab. Theory Related Fields* 108.4 (1997), pp. 517–542. ISSN: 0178-8051,1432-2064. DOI: 10.1007/s004400050119. URL: https://doi.org/10.1007/s004400050119.

[II] Instead of taking the von Neumann algebra generated by the $\{\lambda_s\}_{s\in\Gamma}$, one takes the C^*-algebra generated by them.

[HP00] Fumio Hiai and Dénes Petz. *The semicircle law, free random variables and entropy*. Vol. 77. Mathematical Surveys and Monographs. American Mathematical Society, Providence, RI, 2000, pp. x+376. ISBN: 0-8218-2081-8. DOI: 10.1090/surv/077. URL: https://doi.org/10.1090/surv/077.

[Voi01] Dan Voiculescu. *Free Entropy*. 2001. arXiv: math/0103168 [math.OA]. URL: https://arxiv.org/abs/math/0103168.

[Jun02] Kenley Jung. *A Free Entropy Dimension Lemma*. 2002. arXiv: math/0207149 [math.OA]. URL: https://arxiv.org/abs/math/0207149.

[AGZ10] Greg W. Anderson, Alice Guionnet, and Ofer Zeitouni. *An Introduction to Random Matrices*. Vol. 118. Cambridge Studies in Advanced Mathematics. Cambridge University Press, Cambridge, 2010, pp. xiv+492. ISBN: 978-0-521-19452-5.

[MS17] James A. Mingo and Roland Speicher. *Free Probability and Random Matrices*. Vol. 35. Fields Institute Monographs. Springer, New York; Fields Institute for Research in Mathematical Sciences, Toronto, ON, 2017, pp. xiv+336. ISBN: 978-1-4939-6941-8; 978-1-4939-6942-5. DOI: 10.1007/978-1-4939-6942-5. URL: https://doi.org/10.1007/978-1-4939-6942-5.